Numerical Analysis – Homework #2 (Solutions)

Utah State University Fall 2026

1.

Perform the following calculations using 3-digit rounding, 3-digit chopping and exactly:

(a) $12.3 + 0.0234$

> **Solution:** Exact: 12.3234. 3-digit rounding: 12.3. 3-digit chopping: 12.3.

(b) $-0.0321 + 0.000136$

> **Solution:** Exact: -0.031964. 3-digit rounding: -0.0320. 3-digit chopping: -0.0319.

(c) $12.3 - 0.0234$

> **Solution:** Exact: 12.2766. 3-digit rounding: 12.3. 3-digit chopping: 12.2.

2.

Write the polynomial

$$p(x) = 5x^6 + x^5 + 3x^4 + 3x^3 - x^2 + 1$$

in nested form.

> **Solution:** Coefficients (from x^6 to x^0): 5, 1, 3, 3, -1, 0, 1. Horner form:
>
> $$p(x) = \Big(\Big(\Big(\Big(\big(5x+1\big)x+3\Big)x+3\Big)x-1\Big)x\Big)+1.$$
>
> (Equivalently, show the $+0$ step explicitly if desired.)

3.

Suppose p^* must approximate p with relative error at most 10^{-3}. Find the largest interval in which p^* must lie for each value of p.

(a) $p = 150$

> **Solution:** $|p^* - p| \le 10^{-3}|p| = 0.15 \;\Rightarrow\; p^* \in [149.85,\, 150.15]$.

(b) $p = 1500$

Solution: $|p^* - p| \le 10^{-3}|p| = 1.5 \;\Rightarrow\; p^* \in [1498.5,\, 1501.5]$.

4.

Use 3-digit chopping (after each addition) for

$$\sum_{i=1}^{N} \frac{1}{i}.$$

(a) Find N such that N is the smallest integer for which your sum is not the exact solution (chopped to 3 digits).

Solution: The smallest such N is $N = 7$. With 3-digit chopping after each addition, the running sum gives $S_{\text{chop}} = 2.58$, whereas the exact $H_7 = \sum_{i=1}^{7} \frac{1}{i} = \frac{363}{140} \approx 2.592857$ and $\text{chop}_3(H_7) = 2.59 \neq 2.58$.

(b) What is the absolute error for this value of N?

Solution: Using the exact H_7 as truth:

$$|E| = \left|2.58 - 2.592857142857\right| \approx 0.0128571.$$

(c) What is the relative error for this value of N?

Solution:

$$\text{rel} = \frac{|E|}{|H_7|} \approx \frac{0.0128571}{2.5928571} \approx 4.9587 \times 10^{-3}.$$

5.

The number e can be calculated as

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

Use three-digit chopping arithmetic to compute the following approximations to e. Also compute the relative and absolute errors. For the 'exact' value of e, use $e = 2.7182818$.

(a) $S_1 = \displaystyle\sum_{n=0}^{5} \frac{1}{n!}$

Solution: Evaluate with 3-digit chopping on each term and after each addition (order $n = 0 \to 5$): $S_1 = 2.70$.
Absolute error $|E| = |2.70 - 2.7182818| \approx 1.82818 \times 10^{-2}$.
Relative error $\approx 6.7255 \times 10^{-3}$.

(b) $S_2 = \sum_{n=0}^{5} \frac{1}{(5-n)!}$

Solution: This sums the same terms in reverse order $(1/5!, 1/4!, \ldots, 1/0!)$.
With 3-digit chopping on terms and after each addition: $S_2 = 2.71$.
Absolute error $|E| = |2.71 - 2.7182818| \approx 8.2818 \times 10^{-3}$.
Relative error $\approx 3.0467 \times 10^{-3}$.
(Reversing the order reduces loss of significance.)